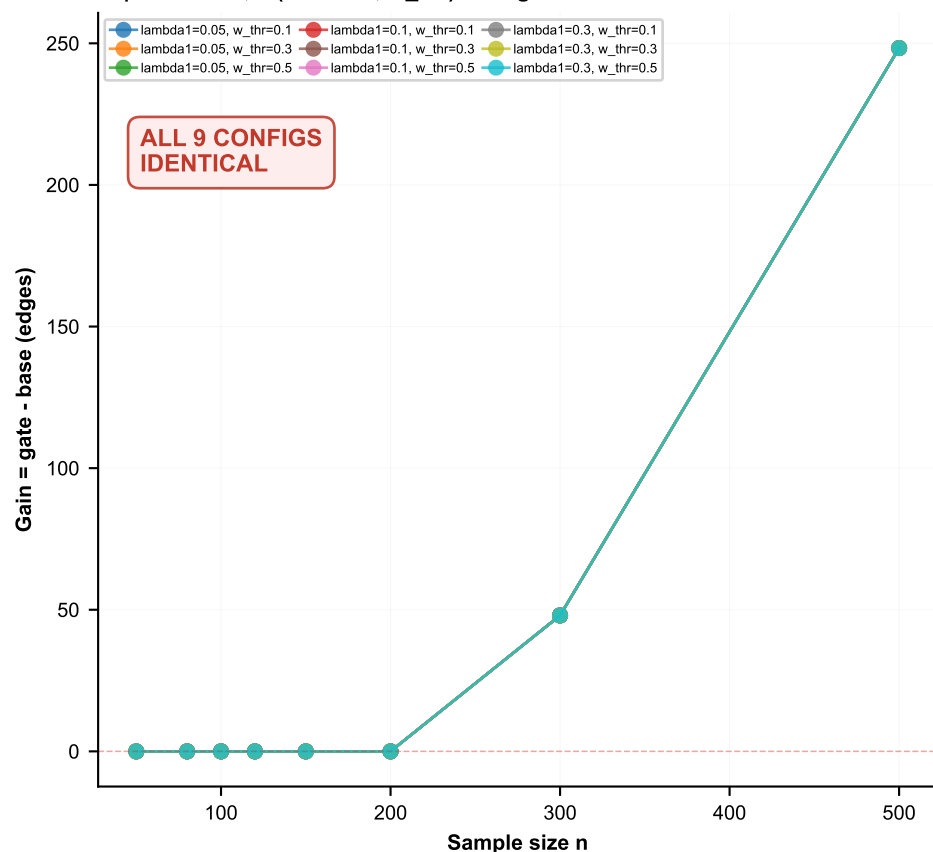
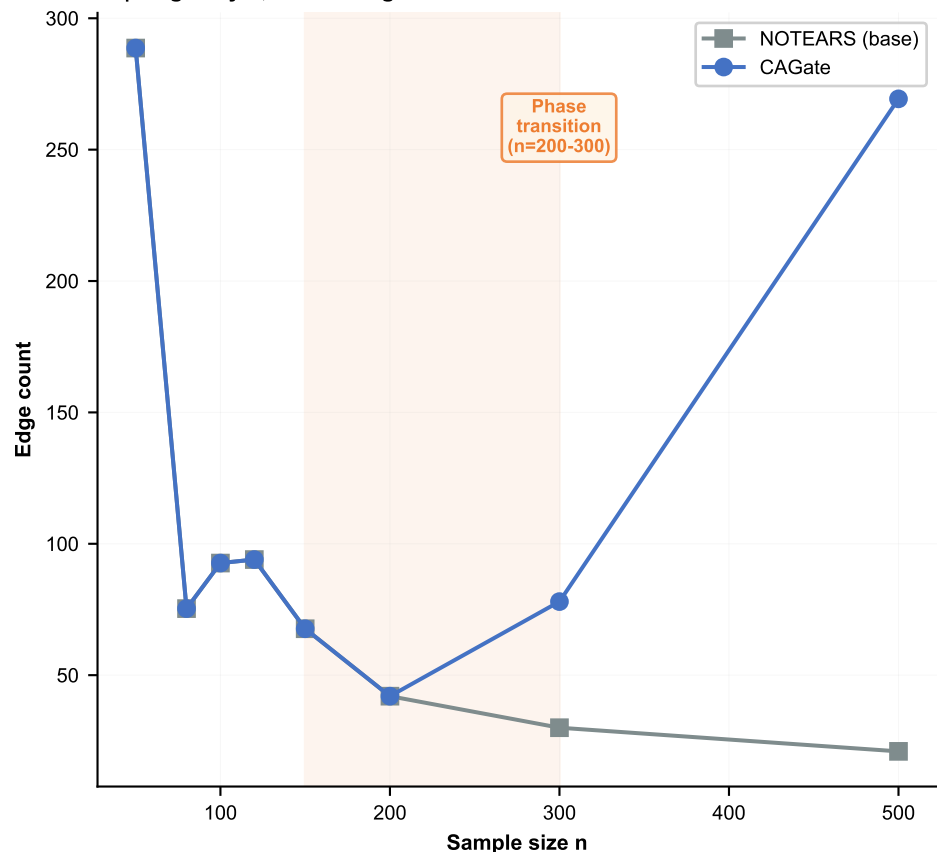


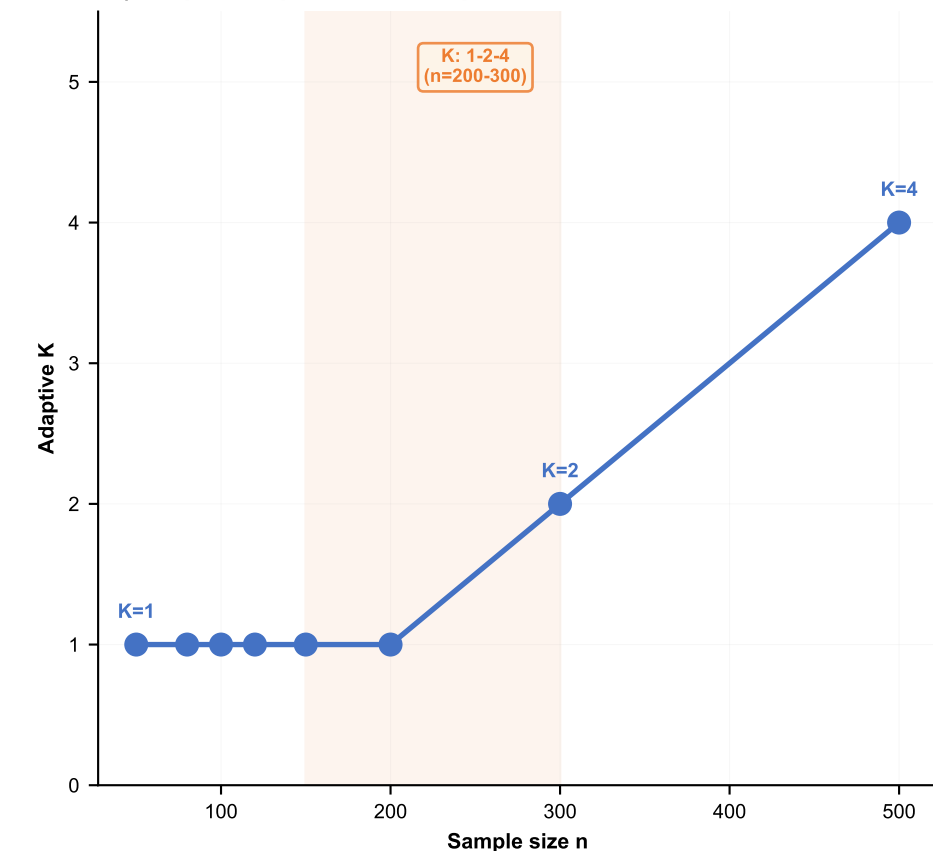
a | Gain vs n, 9 (lambda1, w_thr) configs



b | edges by n, best config



c | Adaptive K - phase transition preserved



d | Summary - defense-grade robustness

Summary: Hyperparameter Sensitivity Analysis

Configurations tested: 9

lambda1 in {0.05, 0.10, 0.30}

w_thr in {0.1, 0.3, 0.5}

Cancer: BRCA, d=100

Mean gain (n < 300): 0 edges

Mean gain (n >= 300): 148 edges

Key findings:

- * All 9 configs produce identical gain(n) curves
- * Phase transition n_crit ~ 200-300 is hyperparameter-invariant
- * Adaptive K preserves transition (not an artifact of fixed K)
- * Conclusion: Phase transition is a data property, not a tuning artifact

DEFENSE-GRADE: PHASE TRANSITION IS HYPERPARAMETER-INVARIANT